AUDREY ROWAN PUTNAM
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EDUCATION

P.h.D Earth, Environmental, and Planetary Sciences, Rice University *May 2026*
Advisor: Dr. Kirsten Siebach
Thesis: "Investigating Geological Evidence for Lava-Water Interaction in Outcrop, Alteration Stoichiometry, and Downstream Sedimentary Detritus."

M.S. Earth Science, Dartmouth College *August 2020*
Advisor: Dr. Marisa Palucis
Thesis: "The Hydrogeomorphic History of Garu Crater: Implications and Constraints on the Timing of Large Late-Stage Lakes in the Gale Crater Region."

B.A. Earth Science, Dartmouth College *June 2018*
Advisor: Dr. Marsia Palucis
Honors Thesis: "Mineralogy, Morphology, and Origin of Martian Layered Deposits: A Global Perspective"

PUBLICATIONS

Peer Reviewed Manuscripts

Putnam, AR, et al. (2026) "Testing the Limits of Provenance Analysis in Mars Analog Coarse Basaltic Fluvial Sediment at Sandvatn, Iceland." *Journal of Geophysical Research: Planets, 131(4), e2025JE009400.*

Putnam, AR, et al. (2024) "Ice-Marginal Volcanic Sequence in Iceland Found on a Nondescript Gradual Hillslope." *Journal of Volcanology and Geothermal Research* 10.1016/j.jvolgeores.2024.108195

Putnam, AR, and MC Palucis. (2021) "The Hydrogeomorphic History of Garu Crater: Implications and Constraints on the Timing of Large Late-Stage Lakes in the Gale Crater Region." *Journal of Geophysical Research: Planets* 126.5: e2020JE006688.

Manuscripts Submitted, In Review, and In Preparation

Putnam, AR, et al. *In Prep.* "Stoichiometric Clustering in Microprobe Measurements of Enigmatic Palagonite". *Submitted to Contributions to Mineralogy and Petrolog*y

ORAL PRESENTATIONS

Putnam, AR, et al. "Stoichiometric Clustering in Microprobe Measurements of the Microcrystalline Mars Analog Material Palagonite" 57th Lunar and Planetary Science Conference, 2026

Putnam, AR, et al. "Selective sorting of basaltic fluvial sand revealed through coordinated mineralogy and geochemistry near Sandvatn, Iceland as a Mars analog" Geological Society of America Connects, 2025

Putnam, AR, et al. “Do Coarse-Grained River Sediments Reflect the Chemistry and Mineralogy of Their Sources in a Mars Analog Watershed in Iceland? Sometimes!” 56th Lunar and Planetary Science Conference, 2025

Putnam, AR, et al. “Ice-marginal lava delta in Iceland found on a nondescript shallow slope: An unexpected record of ice thickness late in deglaciation.” European Geophysical Union Conference 2024, Vienna, Austria No. EGU24-13612. 2024.

Putnam, AR, et al. “Ice-dammed Lake Recorded by Basaltic Lava Deltas above Sandvatn, a Lake in Iceland.” AGU Fall Meeting 2022, Chicago, Illinois No. EP33C-06. 2022

Putnam, AR and Palucis, MC “Stepped-delta record of lake level changes in the Gale region and beyond” Brown University Planetary Lunch Seminar Invited Speaker, Fall 2019

POSTER PRESENTATIONS

Putnam, AR, et al. "Sorting Out Syn-Eruptive Hydrothermal Alteration of Volcaniclastic Rocks in a Mars Analog Basaltic Watershed in Iceland" Tenth International Conference on Mars. No. 3358. 2024.

Putnam, AR, et al. "Characterizing the Basaltic Igneous and Volcaniclastic Provenance at a Mars Analog Site in Iceland with the DIGMARS Team." 53rd Annual Lunar and Planetary Science Conference. No. 1614. 2022.

Putnam, AR and Palucis, MC (2019) “Are Hydrated Silica-Bearing Martian Deltas Evidence of Weathering during Transport or Post-depositional Alteration?” American Geophysical Union Fall Meeting 2019, San Francisco, CA

Putnam, AR and Palucis, MC (2019) “Mineralogy of Martian craters with alluvial fans versus those without: Insights into the controls on Martian alluvial fan distribution,” 50th Lunar and Planetary Science Conference, Woodland Hills, TX

Putnam, AR, Palucis, MC, and Fraeman, AA (2018) “Mineralogy, Morphology, and Origin of Martian Layered Deposits: A Global Perspective,” Wetterhahn Science Symposium presentation and Christopher Reed Senior Honors Thesis competition, May 2018

Putnam, AR and Fraeman, AA (2018) “Mineralogy of Martian layered deposits: A global perspective,” 49th Lunar and Planetary Science Conference, Woodland Hills, TX

Putnam, AR and Kelly, MA (2017). “Mineralogical record of redox conditions in several New England lakes after deglaciation,” Wetterhahn Science Symposium presentation, May 2017

Putnam, AR and Kelly, MA (2016). “Studies on a post-glacial sediment core from Lake Gardner, NH,” Wetterhahn Science Symposium presentation, May 2016

AWARDS, HONORS, & FELLOWSHIPS

Outstanding Graduate Student Award in Earth, Environmental, and Planetary Sciences *2026*
Peter Vail Fellowship in Earth, Environmental, and Planetary Sciences *2022-2023*

- Awarded to a graduate student with high academic records and outstanding qualifications

New Hampshire Space Grant Fellowship *2019*
Caltech Summer Undergraduate Research Fellowship *Summer 2017*

Dartmouth Junior Science Scholar Award *2016-2017*
Dartmouth Sophomore Science Scholar Award *2015-2016*
Rocky Mountain Biological Laboratory independent research grant *Summer 2015*

RESEARCH EXPERIENCE

Reframing Alteration of Basaltic Glass to Palagonite with Cation Stoichiometry, PhD Student, Rice University
2023-2026

- Built dataset of published electron microprobe measurements combined with measurements of our samples
- Reframing data analysis to comparison of measured cation stoichiometry to sidestep disagreement around classification and assumptions that lead to contradictory interpretations

Testing the Limits for Provenance Analysis from Martian Sediments Using an Icelandic Analog Watershed, PhD Student, Rice University *2020-2025*

- Testing recovery of signals of provenance magmatic diversity, eruptive cooling environments, and hydrothermal alteration from fluviodeltaic sand and gravel
- Planned and led the provenance characterization and sampling campaigns for the Summer 2021 and 2022 field seasons
- Integrated data from field observations, optical petrography of thin sections, scanning electron microscope image micro-texture, electron-probe microanalysis data, bulk XRF, ICP-MS, and XRD data, and grain size analysis
- Collected drone imagery to produce digital elevation models via photogrammetry

Expanding the Glaciovolcanic Paleo-Ice Record in Iceland, PhD Student, Rice University *2021-2024*

- Discovered an unexpected volcanic sequence that formed in contact with ice and water and characterized the previously unrecognized style of ice-marginal volcanism
- Added an additional constraint to ice depth and arrangement during a previous deglaciation in Iceland

Constraining the Hydrologic History of the Gale Crater Region, Mars, Through the Geomorphology of Garu Crater, MS Student, Dartmouth College *2018-2020*

- Mapped the geomorphology of a crater in the Gale region, Mars
- Carried out quantitative geomorphic analysis of eroded canyon and sedimentary fan structures
- Applied fluvial morphodynamics to constrain the river discharge and timescale that formed the sedimentary fan
- Utilized crater counting to bound the history of aqueous activity in Garu to younger than 3.5 Ga

Assessing the Correlation of Crater Degradation with Alteration Mineralogy, Mars, MS Student, Dartmouth College *2018-2019*

- Compared statistical distributions of mineral indices from Thermal Emission Spectrometer and multispectral CRISM data to impact crater degradation state across Margaritifer Terra, Terra Sabea, and Tyrrhena Terra, Mars

Mineralogy, Morphology, and Origin of Martian Layered Deposits: A Global Perspective, Senior Honors Thesis, Dartmouth College *2017-2018*

- Expanded the work from my JPL internship with analysis of the geomorphic context of analyzed layered deposits

Analyzed the Mineralogy of Layered Deposits across Mars via Orbital Spectroscopy, NASA Jet Propulsion Laboratory/California Institute of Technology Summer Undergrad Research Fellow *Summer 2017*

- Processed and analyzed hyperspectral visual and near-IR CRISM Mars imagery for over 100 images that overlapped with identified layered deposits to build a global dataset

- Mentored by Dr. Abigail Fraeman

Investigated an Anomalous Layer in New England Postglacial Lake Sediment Cores, Sophomore and Junior Science Scholar, Dartmouth College *2015-2017*

- Measured loss on ignition and magnetic susceptibility along cores of several lakes in the Connecticut river valley
- Identified anomalous characteristics of the layer indicating the transition from a regionally extensive proglacial lake to isolated ponds
- Characterized samples with a scanning electron microscope
- Identified that anomalous magnetic properties were correlated with the presence of pyrite, perhaps indicating a redox control on iron mineralogy
- Mentored by Dr. Meredith Kelly

Monitoring the Effect of Temperature and Precipitation on Mountain White-Crowned Sparrow Nesting Success, Undergraduate Intern, Rocky Mountain Biological Laboratory *Summer 2015*

- Captured, banded, and measured Mountain White Crowned Sparrows in the Rocky Mountains
- Monitored nesting success
- Carried out statistical analysis of prior years' weather data and nesting success
- Mentored by Dr. Ross Conover

TEACHING EXPERIENCE

Organized and Led Discussion for EEPS 604/606 *Spring 2026*

- Organized and taught first-year graduate student journal club
- Organized the internal departmental graduate student talks series. Worked with the graduate student community and faculty to ensure that these talks could be most helpful for students.
- This included organizing the inaugural year of lightning talks by first-year graduate students

Co-organized Departmental Graduate Student Field Trip to Colorado *August 2025*

- Lead for organizing stops and discussed material

Nova Scotia, Canada - Rice University EEPS 591 Type Locality Field Trip *Fall 2024*

- Mentored undergraduate and professional master's students in geological reconnaissance field work and in development of poster presentation content based on their field observations

NSF Planetary Habitability Over Space and Time REU Field Trip in Colorado *Summer 2023*

- Designed and led two days of teaching REU students mostly without geology backgrounds
- Day one introduced students to sedimentology, stratigraphy, and the coevolution of Earth and life with stops at a Cretaceous dinosaur trackway, Jurassic floodplain deposit with exposed bone lags, Permo-Triassic stromatolites, Permian aeolian unit, Pennsylvanian alluvial fan unit and great unconformity at Morrison, CO
- Day two started with stops at other Pennsylvanian outcrops in the Eagle River valley to show students how paleogeography can be constrained. These stops also introduced students to the paleoenvironmental implications of evaporites, volcanic facies at the Holocene Dotsero maar-diatreme volcano, and regional salt tectonic structures.

Guest lecture for EEPS 435/635 Remote Sensing, Rice University *Fall 2022, Spring 2024, Spring 2026*

- Developed and taught a lecture on the use of drones in the geosciences and photogrammetry, including demonstrating aerial mapping with a drone on campus

Guest lecture for EEPS 425/625 Planetary Surface Processes, Rice University *Spring 2025*

- Introduced the basis of sediment sorting, provenance analysis, and volcanic paleo-environmental records as well as the findings of my research to give an example for undergraduates of how class concepts relate to current research on planetary surface processes

Rice University, Earth, Environmental, and Planetary Sciences teaching assistant for EEPS 321 Earth and Planetary Surfaces *Fall 2024*

- Assisted students with field techniques on the field trip to the Panther Branch of Spring Creek
- Helped students learn to use matlab for data processing and modelling in the class laboratory section

Dartmouth College Earth Sciences, teaching assistant for

- EARS 1 How the Earth Works *Fall 2018*
- EARS 40 Materials of the Earth *Summer 2019*
- EARS 19 Habitable Planets *Spring 2019*
- EARS 9 Earth Resources *Winter 2020*
- EARS 3 Elementary Oceanography *Spring 2020*

OUTREACH EXPERIENCE

Rice EEPS Open House *May 2025*

- Lead on assembling, developing, and running a stream table demonstration of fluvial geomorphology for a broad local community audience

Houston ISD Teacher Professional Development *April 2024, January 2025*

- Gave zoom lecture to help teachers engage their students about Mars exploration
- Showed middle school teachers resources like google Mars and discussed strategies for igniting student's curiosity in space exploration

Houston ISD Middle School Planetary Exploration Day *Spring 2023, Fall 2024*

- Ran stations for students to collaboratively race RV rovers through Martian landscapes and work together to learn ballistic trajectories and launch foam space shuttles to specific spots

Rice EEPS Girl Scout Climate Challenge *November 2024*

- Co-led a project allowing students to use infrared thermometers to investigate the urban heat island effect and reflect on strategies to make our built environments more resilient
- Assisted elementary school students with finding fossils in local rocks

Rice/NASA 60th Anniversary of the John F. Kennedy Moon Speech Public Exhibits *September 2022*

- Built a basic stream table to demonstrate how geomorphology can record ancient rivers and lakes in Mars' past
- Explained to the general public the Mars research done at Rice University

Rice Out in STEM Graduate School Panel *October 2024*

- Participated on a panel for LGBTQ+ Rice undergraduate students, sharing advice for navigating graduate school

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MENTORING EXPERIENCE

Current Mentee

Sidikha Tripathee *2025-2026*

- Mentoring first-year through departmental program

Previous Mentees

EEPS Undergraduate *2025*

- Mentoring through the Rice Out in STEM Mentoring program
- Conducting research with another group in the department

Abigail Mebane *Summer 2023*

- NSF Planetary Habitability Over Space and Time REU intern from Pacific Lutheran
- Collected micro-XRF data and supervised classification of scanned thin sections in ENVI
- Was part of the REU field trip to Colorado

Sako Sunami *Spring 2023*

- TOMODACHI program for undergraduate women from Japan and Taiwan from Tohoku University

Marlo Wilcox *2022-2023*

- Assisted in mentoring as she processed drone data via photogrammetry
- Later completed a senior honors thesis and is starting a PhD program to study remote sensing of vegetation

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FIELD EXPERIENCE

Iceland - NASA DIGMARS Project, PhD research *Summer 2021, 2022*

- Characterized and sampled provenance and sedimentary systems at Sandvatn and Apavatn in the Icelandic highlands
- Flew drone mapping surveys over areas of interest
- Assisted collaborators Sarah Simpson and Candice Bedford with day excursions to collect samples from Hengill and Prestanukhur volcanoes

Nova Scotia, Canada - Rice University EEPS 591 Type Locality Field Trip *Fall 2024*

- Produced a poster for the class on Mars implications of the surface expression of curvilinear strata exposed from Triassic fluvial conglomerates of the Wolfville Fm
- Taught on outcrop showing the internal facies and contact between the Triassic North Mountain Basalt flood basalt and contemporaneous aeolian members of the Wolfville Fm
- Observed additional outcrops showing diverse sedimentary environments, bimodal volcanism, salt tectonics, and major faults showing regional structure.
- Connected with local indigenous museum paleontologists to learn about the intersection of regional geology with the history of the Mi’kmaq First Nation, European settlement, and early geologic research by Charles Lyell

Colorado - NSF Planetary Habitability Over Space and Time REU Field Trip *Summer 2023*

Virginia - Rice EEPS Graduate Student Field Trip *Spring 2023*

Book Cliffs and San Rafael Swell, Utah - Rice University EEPS Siliciclastic Depositional Systems Field Trip *Spring 2023*

Arkansas - Rice University EEPS Siliciclastic Depositional Systems Field Trip *Spring 2022*

New England Regional Intercollegiate Geological Conference day trips *Fall 2018*

- Day trips on Northern Vermont thrust faulting, Hudson valley proglacial lakes, and a western Massachusetts suture between paleozoic terranes

Alberta, Montana, Wyoming, Utah, Nevada, California, Arizona - Dartmouth College Earth Science 'Stretch' Undergraduate Off Campus Program *Fall 2016*

- 10 week field trip with seven sections taught by different department professors
- Diverse field experiences including field mapping of sedimentary and volcanic areas, radar echo sounding on the Athabasca glacier, measuring solutes in Colorado river,